

Internship Report

Of

Web Development

"ShopEase – A Responsive E-Commerce Website"

Submitted

To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of
B.Tech in Computer Science and Engineering



By

Roll Number of Student: 25SCS1003003494

Name of Student: Sachin Kumar Gupta

Batch: 2025–2029 | Semester: 3rd | Section: 2CSE_9

CANDIDATE'S DECLARATION

I, Sachin Kumar Gupta, do hereby declare that the internship report titled “ShopEase – A Responsive E-Commerce Website” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references used have been duly acknowledged, and I have not copied or taken material from any other source without proper citation. I affirm that this report represents my own work carried out during my internship and has not been submitted, in full or in part, to any other institute for any degree or diploma requirement. I shall be solely responsible for any kind of copyright violation in this regard, if found otherwise.

Signature

Student Name: Sachin Kumar Gupta

Roll No.: 25SCS1003003494

ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to everyone who supported and guided me throughout the course of my internship. I am thankful for their patient mentorship, constructive feedback, and encouragement, which helped me apply my classroom learning to a practical, real-world project.

I am sincerely grateful to Codec Technologies Pvt. Ltd. and the National Internship Portal for giving me the opportunity to undertake this Web Development internship, and to my mentors for sharing their technical insights and guidance throughout the program.

I would also like to thank the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous academic support and encouragement during this internship.

Finally, I express my heartfelt gratitude to my parents and family for their constant support and blessings.

Signature

Student Name: Sachin Kumar Gupta

Roll No.: 25SCS1003003494

INTERNSHIP COMPLETION CERTIFICATE



TABLE OF CONTENTS

Chapter No.	Chapter Name	Page Nos.
--	Cover Page	--
1.	Candidate's Declaration	i
2.	Acknowledgement	ii
3.	Internship Completion Certificate	iii
4.	Project Description 4.1 Introduction 4.2 Organization Profile 4.3 Problem Statement 4.4 Project Objectives 4.5 Scope of the Project 4.6 Technologies and Tools Used 4.7 System Architecture 4.8 Methodology 4.9 Expected Outcomes 4.10 Certificate of Completion & Communication Proof	1 – 4
5.	Bibliography / References	5

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents the work carried out during a one-month Web Development internship undertaken with Codec Technologies Pvt. Ltd. under the AICTE & ICAC approved internship program hosted on the National Internship Portal. The internship focused on strengthening practical, hands-on skills in front-end web technologies and applying them to build a functional, responsive web application. As part of this internship, a project titled “ShopEase – A Responsive E-Commerce Website” was designed and developed, covering the complete workflow of planning, designing, coding, testing and refining a real-world style web application.

The internship provided practical exposure to industry-relevant tools and practices, including version control using Git and GitHub, responsive web design principles, and client-side scripting using modern JavaScript. This chapter describes the organization, the problem addressed by the project, its objectives, scope, the technologies used, the system architecture, the methodology followed, and the outcomes achieved.

4.2 Organization Profile

Codec Technologies Pvt. Ltd. is a skill-development and ed-tech organization that partners with the National Internship Portal – an initiative associated with the Ministry of Education – to offer AICTE & ICAC approved internship programs to engineering and technology students across India. Working in association with Google for Education, the organization conducts structured, mentor-guided internship tracks in domains such as Web Development, Data Science, Machine Learning and Cybersecurity.

The Web Developer Internship track is designed to help students transition from theoretical, classroom-based learning to practical, project-based learning by exposing them to real development workflows, coding best practices, and modern front-end tools within a short, intensive one-month duration.

4.3 Problem Statement

Small and first-time online sellers often find it difficult to showcase their products on the web without investing in a heavy, expensive e-commerce platform. Many beginner-friendly templates available online are either not responsive across devices, lack a working shopping cart, or do not allow visitors to search and filter products efficiently. There is a genuine need for a lightweight, responsive e-commerce front-end that lets a visitor browse products, search and filter them, and add them to a cart – with the cart data persisting even after the page is refreshed – without depending on a heavy back-end infrastructure.

4.4 Project Objectives

- To design a clean, responsive, multi-page e-commerce website that works seamlessly across desktop, tablet and mobile screen sizes.
- To implement a dynamic product listing page with category-based filtering and keyword-based search.

- To build a fully functional shopping cart that supports adding, updating quantity, and removing items.
- To persist cart data across page reloads using the browser's localStorage.
- To validate and handle a working contact/enquiry form on the client side.
- To follow good coding practices, semantic HTML structure, and maintain the project using Git/GitHub version control.

4.5 Scope of the Project

The scope of the project is limited to the front-end of an e-commerce website. It includes a home page, a product listing page with filter and search functionality, a product detail view, a shopping cart page, and a contact page. The project uses static/local JSON data to simulate a product catalogue rather than connecting to a live back-end server or a payment gateway, since the focus of the one-month internship was on front-end development, responsive design and client-side logic. Integration with a real back-end, database and payment gateway is identified as a natural extension of this project beyond the internship duration.

4.6 Technologies and Tools Used

The following table lists the major technologies and tools used to build the project.

Category	Technology / Tool	Purpose
Markup & Styling	HTML5, CSS3, Bootstrap 5	Page structure and responsive UI design
Scripting	JavaScript (ES6)	Cart logic, search/filter, form validation
Data	JSON	Local product catalogue
Storage	Browser localStorage	Persisting cart data across sessions
Version Control	Git & GitHub	Source code management and versioning
Tools	VS Code, Chrome DevTools	Development, debugging and responsive testing

Table 4.1: Technologies and Tools Used

4.7 System Architecture

The project follows a simple client-side layered architecture. The Presentation Layer (HTML5, CSS3 and Bootstrap 5) renders the responsive user interface. The Application Logic Layer, written in JavaScript (ES6), handles cart operations, product filtering/search and form validation. The Data Layer supplies product information from a local JSON file, while the Browser Storage layer (localStorage) persists the shopping cart between sessions. Figure 4.1 below illustrates this architecture.

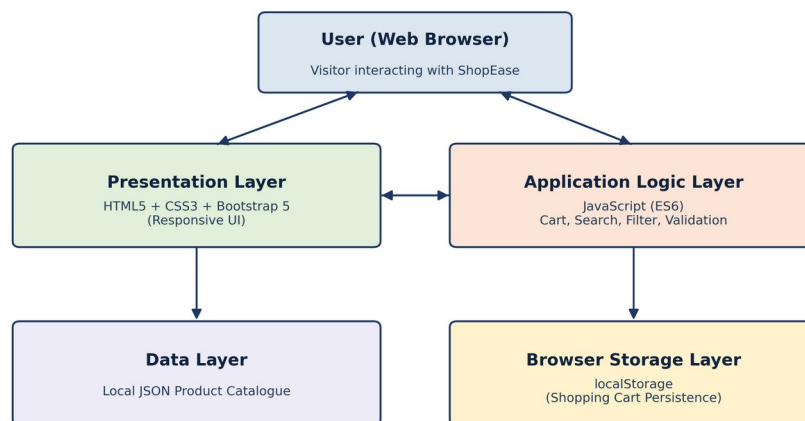


Figure 4.1: System Architecture of ShopEase

4.8 Methodology

The project was developed in the following stages:

- Requirement Analysis – Identifying the core pages and features needed for a basic e-commerce experience.
- Design – Creating wireframes and deciding the layout, colour scheme and navigation flow for all pages.
- Development – Building the HTML structure, styling with CSS/Bootstrap, and implementing JavaScript logic for the cart, search and filters.
- Testing – Cross-checking responsiveness on different screen sizes and validating cart and form behaviour.
- Refinement & Deployment – Fixing bugs, optimizing code, and pushing the final version to GitHub.

4.9 Expected Outcomes

- A fully responsive, multi-page e-commerce website front-end that works across devices.
- A working shopping cart with add, update and remove functionality, persisted via localStorage.
- A searchable and filterable product listing page.
- Improved practical proficiency in HTML5, CSS3, Bootstrap and JavaScript (ES6).
- Hands-on experience with Git/GitHub-based version control and a structured development workflow.

4.10 Certificate of Completion and Other Communication Proof

The internship completion certificate issued by Codec Technologies Pvt. Ltd. through the National Internship Portal is attached in Chapter 3 of this report, and is reproduced below for reference along with the program details.



Figure 4.2: Internship Completion Certificate – Sachin Kumar Gupta (Web Developer Intern, Codec Technologies Pvt. Ltd.)

5. BIBLIOGRAPHY / REFERENCES

- [1] Codec Technologies Pvt. Ltd. – National Internship Portal, Internship Completion Certificate, 2026.
- [2] Mozilla Developer Network (MDN Web Docs) – HTML, CSS and JavaScript documentation, <https://developer.mozilla.org>
- [3] W3Schools – Web development tutorials and references, <https://www.w3schools.com>
- [4] Bootstrap Official Documentation, <https://getbootstrap.com/docs>
- [5] GitHub Docs – Version control and collaboration, <https://docs.github.com>
- [6] freeCodeCamp – Responsive Web Design and JavaScript curriculum, <https://www.freecodecamp.org>